ComfortRoute CSE 3311 – Iteration 3

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GitHub Repository: https://github.com/Kass-ii/CSE-3311-

Version: iteration 0.3

LIVE DEMO

Iteration 3 Goals

- Focuses on supporting origin-to-destination trips while minimizing idle time.
- Implementing a flexible routing structure while improving map interaction.
- App customization and account settings for user preference and updating personal information.

Origin-Destination Routing

ComfortRoute

Find a full route between two stations using comfort-based routing.

Try a station like "centreport" or "downtown".

Origin Station

CENTREPORT STATION

Destination Station

TRINITY LAKES STATION

Departure Time

10:00:00 AM

Best Route Found

ComfortRoute found a route on CENTREPORT STATION → Bell Station → TRINITY LAKES STATION that keeps you on the train for 19 minutes. You arrive at 10:22 AM.

Route Summary

Route: CENTREPORT STATION → Bell Station → TRINITY LAKES STATION

Total Time on Train: 19 minutes

Total Wait Time: 3 minutes

Leg 1

From: CENTREPORT STATION

To: Bell Station

Depart: 10:00 AM

Arrive: 10:07 AM

Route: TRE

Leg 2

From: Bell Station

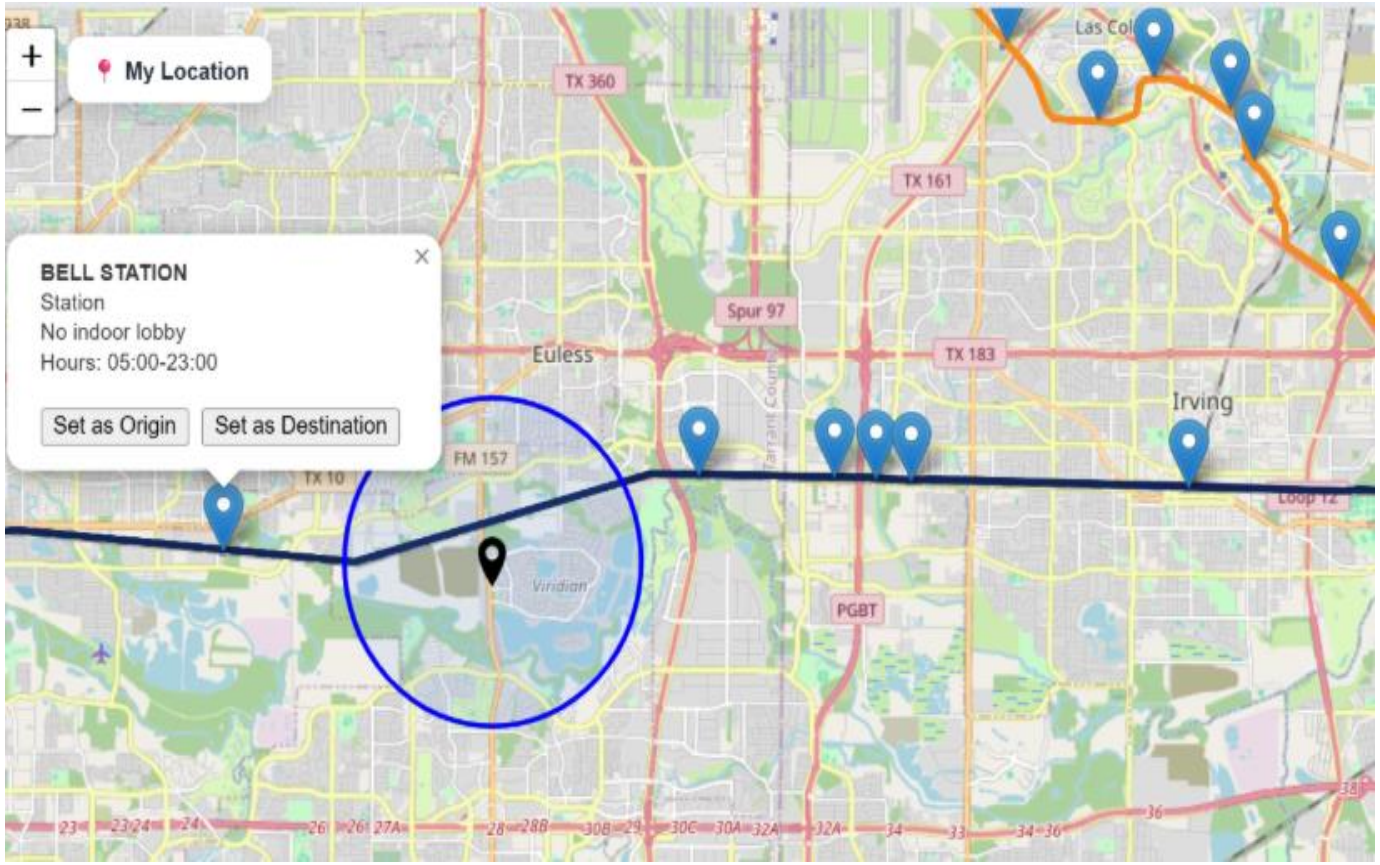
To: TRINITY LAKES STATION

Depart: 10:10 AM

Arrive: 10:22 AM

Route: DART Green

Map Interactivity

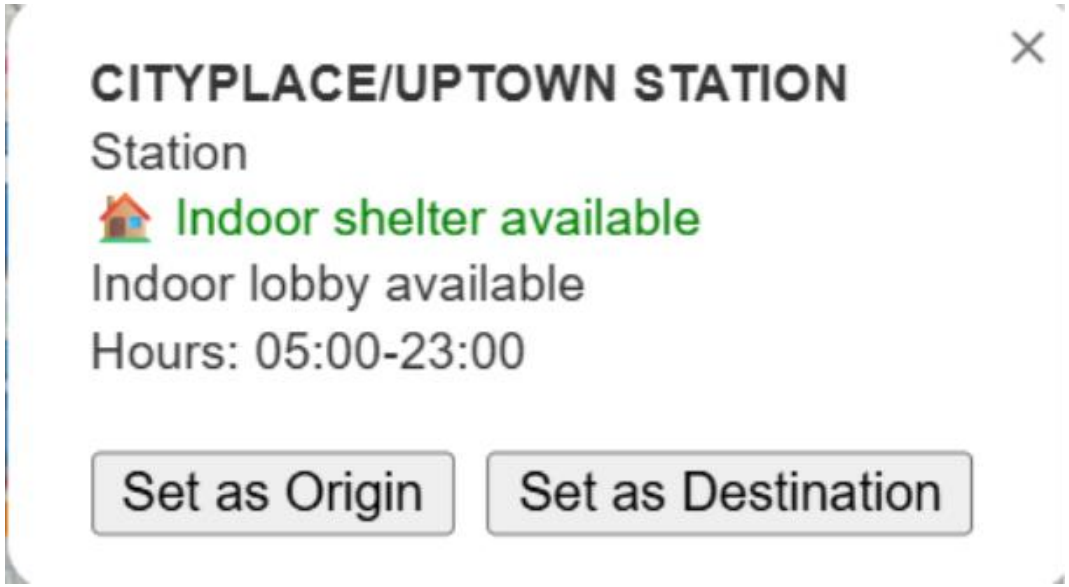


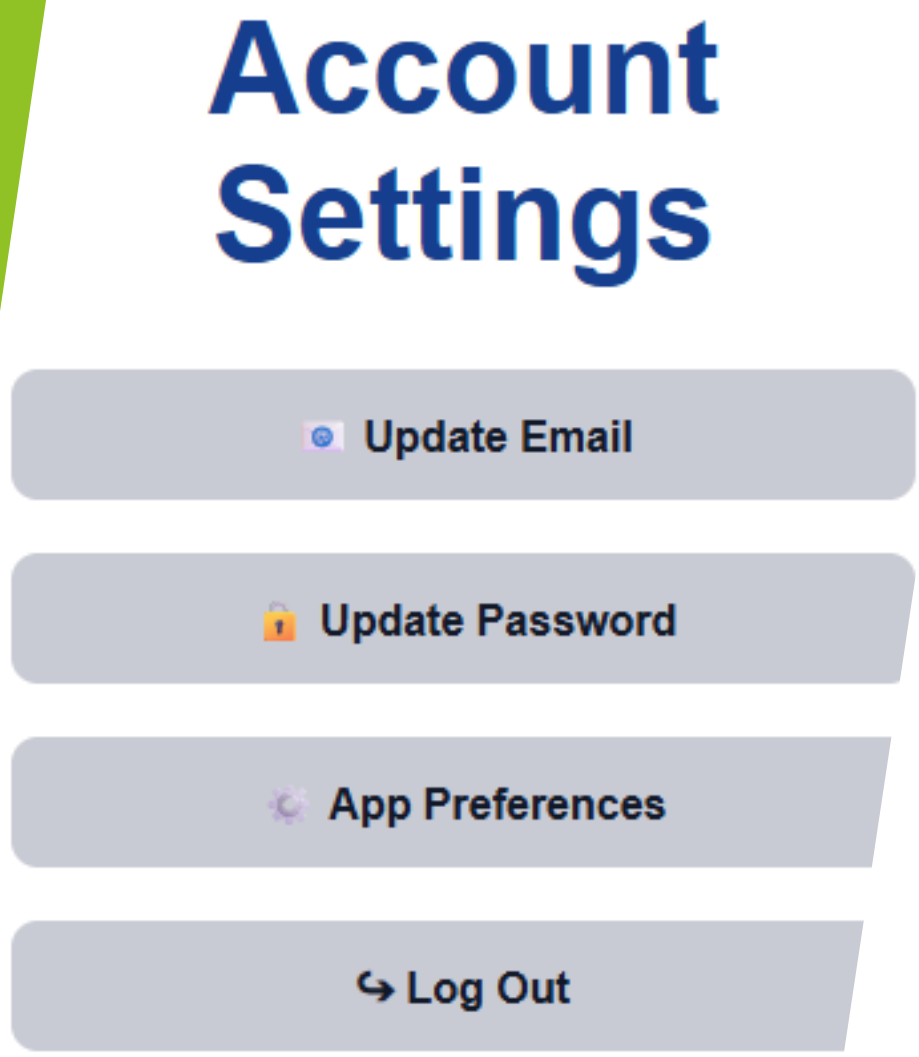
- Allows the selection of origin and destination stations through map interaction
- Displays nearby stations that are relative to the user's current location.
- Specific transit lines are distinguished by color.

Comfort Features

- When pulling up station information, users will be able to view station operation and indoor lobby hours.

- Gives users insight on which stations maximizes personal comfortability.





Personalized Settings

- Users may update email and password information within the app.
- App preferences that give the user the option to highlight station details (operating hours, indoor lobbies, etc.)
- Log-out system.

Risk

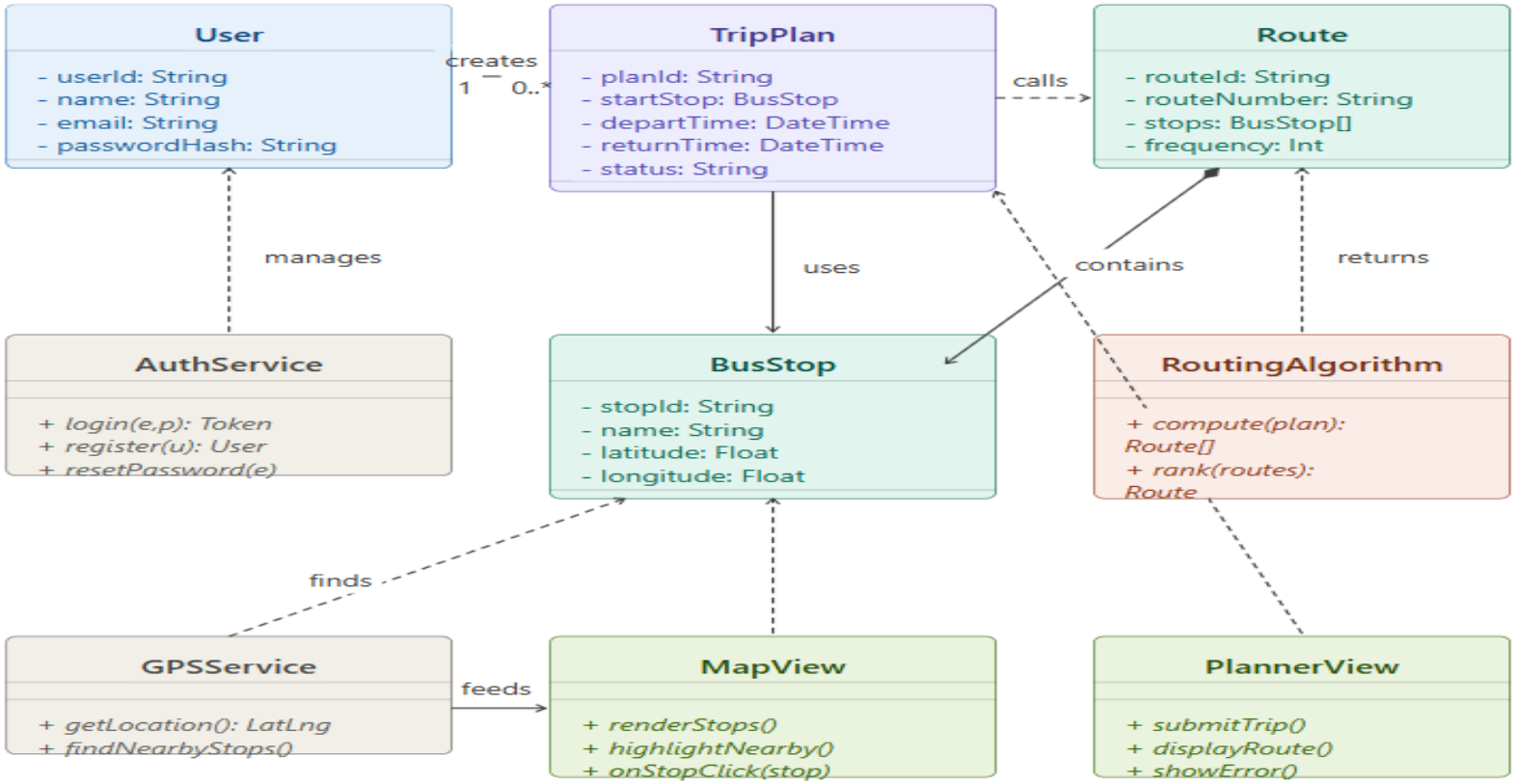
- Make transfer time user configurable
- Use old raspberry pi as stopgap server
- Display service alerts to users, filter alerts for those relevant to the user's trip and warn if route is affected.

Risk	Time Impact	Probability	Risk Exposure
(1) Train platforms are far apart and transfers take longer than expected.	4 hours	10%	24 min
(2) Unable to deploy app to host	6 hours	20%	2 hours
(3) GTFS schedules do not reflect reality on transit	4 hours	60%	2.5 hours

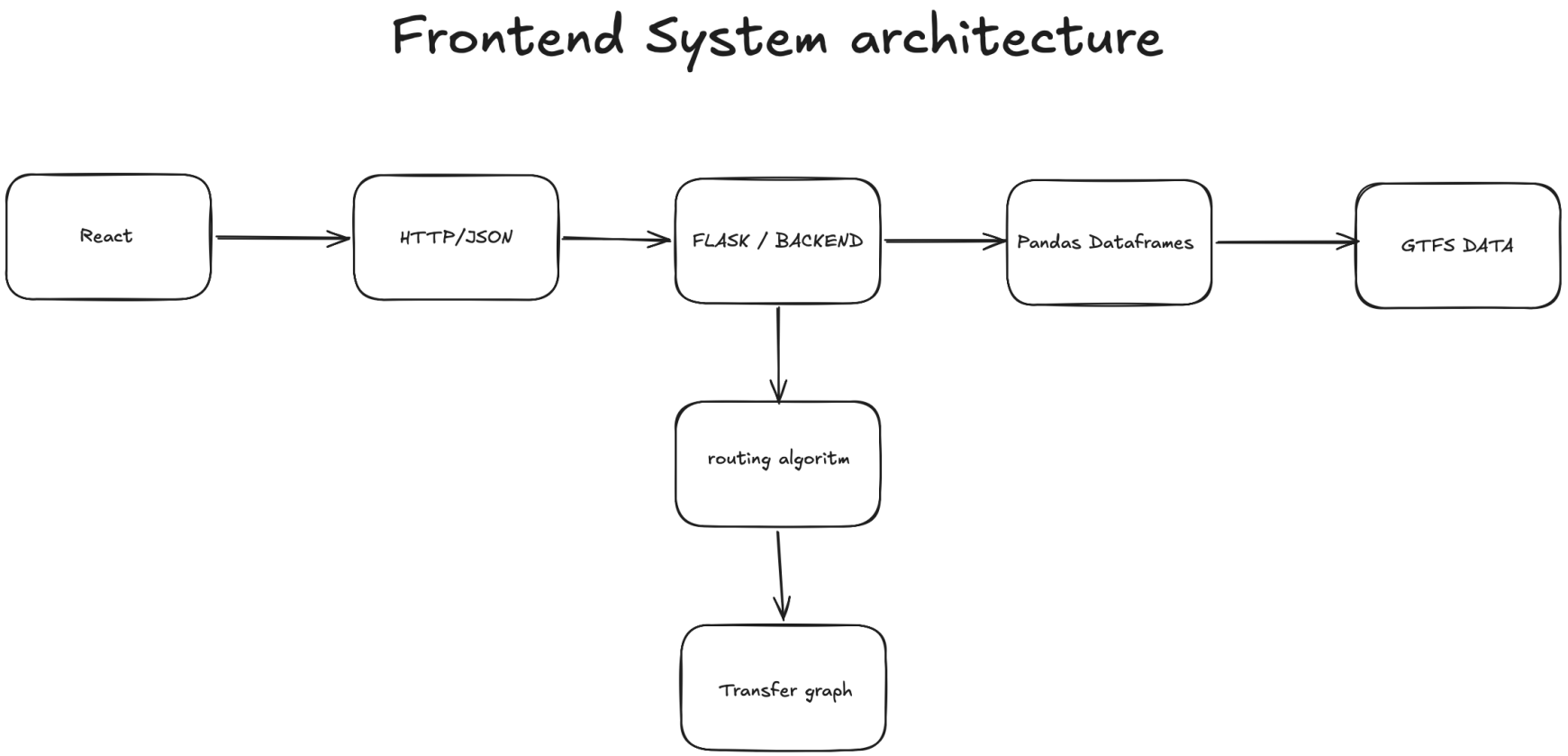
User feedback

- Gave demos to friends who use transit frequently and asked for feedback for future iterations
- Will be implemented for iteration 4
- Mainly UI/UX improvements:
  - bookmark favorite stations and save to user account
  - show multiple trip options and compare comfortable trip with the fastest
  - configurable buffer time between transfers
  - A now button next to the departure time, replace -- with 00 if user leaves blank
  - Display DART service alerts to user if they affect the route
  - Dropdown menu to autocomplete station names in the planner

UML Class Diagram



System Architecture: Frontend



Roadmap / Next Iteration

- Must have:
  - Configure and deploy app in production environment
  - Visual representation of suggested trip
  - Multi line transfers
- Should have:
  - Bookmark frequently used stations
  - Configurable transfer buffer
- Could have:
  - Scrape DART service alerts
  - Multiple trip options to compare
- end

Backtracking and Deployment

- Explores all possible solutions recursively
  - Tries one choice at a time and checks if it is valid
- Continues deeper only when the current path works
  - Reverses the last choice when a path fails
- Repeats until a valid solution is found or all options are tested
- Deployment infrastructure

THANK YOU!!!!
ANY QUESTIONS?????